

Krishnaa Vadivel | Electrical Engineer

| Semiconductor Devices | Photonics | FPGA Systems

📞 516-853-5663 • ✉️ srikrishnaa.careers@gmail.com • in sri-krishnaa-ece-phd
GitHub: krishnaa423 | LeetCode: krishnaa42342 | Docker Hub: krishnaa42342

Summary

Electrical engineering PhD researcher with Cornell and Yale ECE training across semiconductor physics, embedded systems, photonics, and accelerator-enabled scientific computing. Experience spans FPGA and MCU development, CUDA/OpenGL engineering tools, semiconductor-device modeling, and digital-design projects in Verilog.

Experience

Yale University, Electrical Engineering

PhD Research Assistant / Embedded Systems TA

2021–present

- Conduct research at the intersection of semiconductor physics, condensed-matter modeling, and high-performance scientific computing, with supporting coursework and theory development in semiconductor devices, photonics, and signals/systems.
- Teach and support embedded systems instruction, reinforcing firmware, instrumentation, debugging, and system-integration skills relevant to practical EE development.
- Build numerical and software workflows for device-scale and materials-scale simulation using Python, C++, HPC toolchains, and accelerator-enabled platforms.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state materials physics.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed embedded firmware and FPGA-based sensor-logging functionality for downhole geological tools using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-class platforms.
- Built CUDA/OpenGL software for large acoustic waveform processing, engineering visualization, and diagnostic analysis in field-tool workflows.
- Supported circuit and system bring-up with oscilloscope-based validation, field testing, and targeted PCB updates in Altium.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content on lasers, integrated optics, and optoelectronic components for engineering-facing audiences.

Selected Projects

Biosignal Robotic Hand: Published Cornell ECE project combining surface EMG sensing, analog filtering, PIC32 control logic, and servo actuation for robotic finger motion.

FPGA AI Accelerator: Small Verilog matrix-multiply accelerator with simulation-backed verification for ML-oriented datapath design.

Semiconductor Simulation: Numerical modeling of semiconductor electrostatics, drift-diffusion transport, and pn-junction-style device behavior.

Integrated Photonics Modeling: FDTD simulation of resonator and band-pass style photonic structures using Meep.

Digital Design: Single-cycle MIPS processor in Verilog with self-checking test bench and documentation.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Hardware: Verilog, FPGA, MCU firmware, PCB bring-up, instrumentation, oscilloscope-based debugging

Devices: Semiconductor physics, pn junctions, MOSFET concepts, transport modeling, thin films

Photonics: Integrated photonics, lasers, optics, FDTD simulation

Software: Python, C, C++, CUDA, OpenGL, Linux

Systems: Docker, Kubernetes, scientific computing, HPC workflows